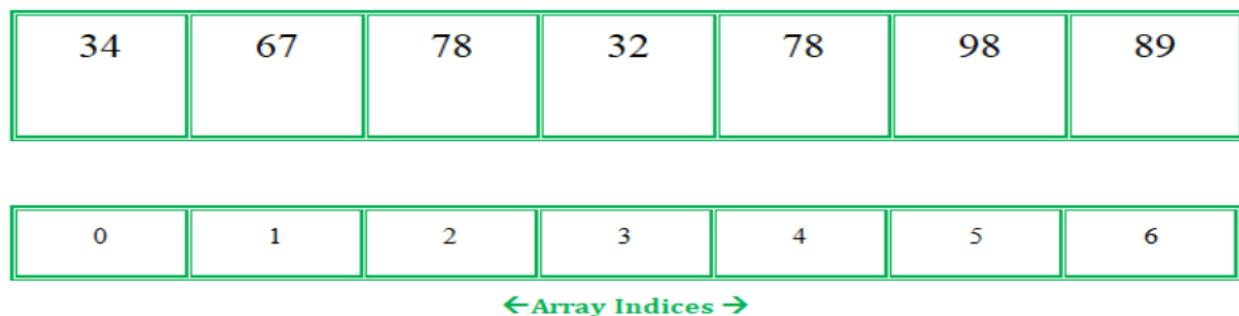


8-1 Array of Objects

An array in C/C++ or be it in any programming language is a collection of similar data items stored at contiguous memory locations and elements can be accessed randomly using indices of an array. They can be used to store the collection of primitive data types such as int, float, double, char, etc of any particular type. To add to it, an array in C/C++ can store derived data types such as structures, pointers, etc. Given below is the picture representation of an array.

Example:

Let's consider an example of taking random integers from the user.



Array of Objects

When a class is defined, only the specification for the object is defined; no memory or storage is allocated. To use the data and access functions defined in the class, you need to create objects.

Syntax:

```
ClassName ObjectName[number of objects];
```

✚ The Array of Objects stores *objects*.

✚ An array of a class type is also known as an array of objects.

Example1:

Storing more than one Employee data. Let's assume there is an array of objects for storing employee data emp[50].

Objects	Employee Id	Employee name
emp[0]→		
emp[1]→		
emp[2]→		
emp[3]→		

Below is the C++ program for storing data of one Employee:

```
#include<iostream>
using namespace std;

class Employee
{
    int id;
    char name[30];    يحدد عدد الحروف المسموحة للاسم
public:
    void getdata();//Declaration of function
    void putdata();//Declaration of function
};

void Employee::getdata(){//Defining of function
    cout<<"Enter Id : ";
    cin>>id;
    cout<<"Enter Name : ";
    cin>>name;
}

void Employee::putdata(){//Defining of function
    cout<<id<<" ";
    cout<<name<<" ";
    cout<<endl;
}

int main(){
    Employee emp; //One member
    emp.getdata();//Accessing the function
    emp.putdata();//Accessing the function
    return 0;
}
```

Output:

```
Enter Id : 1
Enter Name : Ali
1 Ali
```

This program can take the data of only one Employee. What if there is a requirement to add data of more than one Employee. Here comes the answer **Array of Objects**. An array of objects can be used if there is a need to store data of more than one employee. Below is the C++ program to implement the above approach.

```
#include<iostream>
using namespace std;

class Employee
{
    int id;
    char name[20];    يحدد عدد الحروف المسموحة للاسم
public:

    // Declaration of function
    void getdata();

    // Declaration of function
    void putdata();
};

// Defining the function outside the class

void Employee::getdata()
{
    cout << "Enter Id : ";
    cin >> id;
    cout << "Enter Name : ";
    cin >> name;
}

// Defining the function outside the class

void Employee::putdata()
{
    cout << id << " ";
    cout << name << " ";
    cout << endl;
}

// Driver code
int main()
{
    // This is an array of objects having
    // maximum limit of 30 Employees
    Employee emp[30];    مصفوفة الكائنات تتسع لمليء معلومات 30 موظف
    int n, i;
    cout << "Enter Number of Employees - ";
```

```

cin >> n;

// Accessing the function
for(i = 0; i < n; i++)
    emp[i].getdata();

cout << "Employee Data - " << endl;

// Accessing the function
for(i = 0; i < n; i++)
    emp[i].putdata();
}

```

Output:

```

Enter Number of Employees - 3
Enter Id : 1
Enter Name : ali
Enter Id : 2
Enter Name : zainab
Enter Id : 3
Enter Name : hassan
Employee Data -
1 ali
2 zainab
3 hassan

```

In this example, more than one Employee's details with an Employee id and name can be stored.

- Employee emp[30] - This is an array of objects having a maximum limit of 30 Employees.
- Two for loops are being used-
 - First one to take the input from user by calling emp[i].getdata() function.
 - Second one to print the data of Employee by calling the function emp[i].putdata() function.

Example :

Create object array to input the name of item and its price for Sales store.

Sol.

```

#include<iostream>
using namespace std;
class item
{
    char name[30];
    int price;
public:
    void getitem();
    void printitem();
};

// Function to get item details
void item::getitem()
{
    cout << "Item Name = ";
    cin >> name;
}

```

```
    cout << "Price = ";
    cin >> price;
}

// Function to print item
// details
void item :: printitem()
{
    cout << "Name : " << name <<
        "\n";
    cout << "Price : " << price <<
        "\n";
}

const int size = 3;

// Driver code
int main()
{
    item t[size];
    for(int i = 0; i < size; i++)
    {
        cout << "Item : " <<
            (i + 1) << "\n";
        t[i].getitem();
    }

    for(int i = 0; i < size; i++)
    {
        cout << "Item Details : " <<
            (i + 1) << "\n";
        t[i].printitem();
    }
}
```